

Security and Protection of Systems

Lecture 05 *Wireless & Mobile Systems Security*

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Outlines

- Bluetooth Systems
- Near Field Communication Systems
- Radio Frequency Identification Systems
- WiFi Systems
- Mobile Systems
- Security Solutions

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Bluetooth Systems

- Bluetooth systems are applications that use Bluetooth technology that depends on short range radio frequency transmission.
- Examples: Wireless Mouse - Wireless Keyboard - PAN - BAN.
- Piconet is established when two devices come within range each others.
- **Master:** controlling all wireless traffic.
- **Slave:** Receive commands
- 1. **Active Slave:** Scanning transmissions
- 2. **Parked Slave:** Connecting without participation

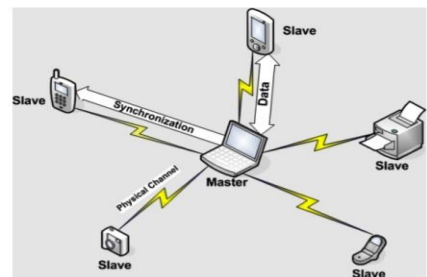


Fig. PICONET

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Bluetooth Attacks

1- Bluejacking Attack:

- Sends undesirable packets to Bluetooth enabled devices. (text - image - sound - video)
- Consider more annoying than harmful
- No data stolen



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2- Bluesnarfing Attack:

- Accessed unauthorized information from wireless device through a Bluetooth connection
- Copies information without owner knowledge such as coping emails and contact.

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Near Field Communications

- NFC is a technology used to establish communication between devices in close distance.
- Once two devices close to each other in the range less than 20 cm, can read information as well as transmit data.



Near Field Communications Attacks

1- Eavesdropping NFC:

- An unencrypted NFC communication can be intercepted and viewed
- Attacker must be extremely close to pick the NFC communication Therefore, users should remain aware of their surrounding while making services such as payments.



2- MitM NFC:

An attacker can intercept the NFC communication between devices and forge a facilitate response.

Note: Devices can be configured pairing so one device can only send and the other can only receive..

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Radio Frequency Identification (RFID)

- Commonly used to transmit information between employee identification badges, inventory tags, book labels, and other paper-based tags that can be detected by a proximity reader.
- Most RFID are passive
 - Because, it is a very small and did not require a power supply.
- RFID tags are susceptible to different attacks
- Contains some security enhancements over the previous version

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RFID Attacks

1- Unauthorized tag access

- A Rogue RFID reader can determine the inventory on a store shelf to track the sales of specific items.
- Sales information could be used by a rival product manufactures to negotiate additional shelf space of better product placement.



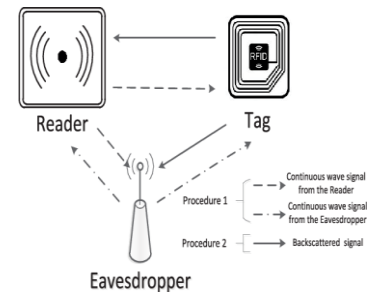
2- Fake tags

- Fake tags with some fictitious information are replaced instead of authenticated tags.
- Fictitious information that are given by fake tags gives illusion to user about product that not exist in the store



3- Eavesdropping Tags:

- Unauthorized user could track on transmission between RDIF user and readers.
- Reveal secret information to rivals



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WiFi Systems

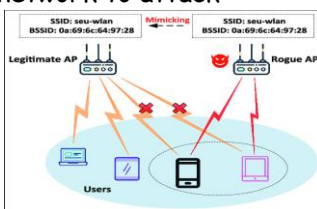
- Commonly used for local area networking of devices and Internet access, allowing nearby digital devices to exchange data by radio waves.
- Most WiFi attacks are
 1. Rogue access point
 2. Evil twins
 3. Intercepting wireless data



WiFi Attacks

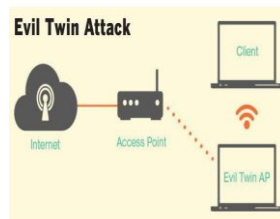
1- Rogue access point

- An unauthorized access point that allows an attacker to bypass network security configuration
- Usually setup by an insider (Employee)
- May be setup behind a firewall, opening the network to attack



2- Evil twins

- Access point setup by an attacker
- Attempts to mimic an authorized access point
- Attackers capture transmissions from users to evil twins access point



3- Intercepting wireless transmission

- An attacker can pick up the Radio frequency signal from an open misconfigured access point
- Using WiFi to read this data could yield significant information to an attacker regarding the wired enterprise network.
- Could be Replay, MitM, DoS attack

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Mobile devices

- **Types:** Tablet: Smartphones, Wearable, Laptop, Notebook
- **Connectivity Methods:** Cellular, Satellite, Infrared, ANT, USB connections
 - ANT- A propriotor wireless network technology used by sensors for communicating data
- Mobile devices may faces different types of security risks such as :
 1. **Mobile device vulnerabilities**
 2. **Connection vulnerabilities**

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Mobile Device Vulnerabilities (1)

- **Physical Security:** A result of the loss or theft of mobile devices
- **Limited firmware update:** Apple iOS is closed and propriety architecture, Google does not create the hardware that Android run on.
- **Location Tracking:** Mobile devices with GPS capabilities support geolocation, Attacker can determine where user are and plan to steal information or inflict harm
- **Unauthorized Recording:** Mobile activated could be recorded during the service.

Connection Vulnerabilities (2)

- **Tethering:** Mobile device with an active internet connection can be used, therefore, an unauthorized mobile device may infect other tethered mobile devices or the corporate network.
- **USB in-the-Go (OTG):** Mobile device with a USB connection can act as either a host or a peripheral used for external media access, therefore, an infected computer could allow malware to be sent to the device.
- **Connecting to the public network:** Mobile devices must at time use public external networks for internet access, therefore, attacker can eavesdropping data transmission and view sensitive information

Securing Mobile Devices Considerations

1st Consideration: Configuring the device:

1. Disable unused feature: disable Bluetooth wireless data communication in order to prevent bluejacking and bluesnarfing.
2. Use strong authentication: Restrict unauthenticated users with a screen lock and require a strong passcode.

Securing Mobile Devices Considerations

2nd Consideration: Configuring device app security:

Significant loopholes in which mobile device data can be accessed through:

1. 1- Data-in-transit: carrier build surveillance capabilities into their networks, allows law enforcement agencies to collect data-in-transit, new mobile apps over the -the-top content
2. 2-Remote- data-at-Rest: Apple and google possess decryption key necessary to unlock data on their services. Courts routinely server order to apple and google to provide data stored on their servers, Users can choose to turn off backups to iCloud or Google servers.

Securing Mobile Devices Considerations

3rd Consideration: Using mobile management tools:

Storage segmentation: Separating business data from personal data

- Containerization: Separating storage into containers and managing each appropriately.
- Advantages : Helps companies avoid data ownership privacy issues and legal concern regarding a user's personal data, and allows companies to delete only business data when necessary without touching personal data.

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WiFi Protected Access (WPA)

- Could be implemented through firmware upgrades on wireless network interface cards.
- Implements the Temporal Key Integrity Protocol (TKIP).
 - Employs a per-packet key, meaning that it dynamically generates a new 128-bit key for each packet and thus prevents the types of attacks.
- Includes a Message Integrity Check (CRC), which is designed to prevent an attacker from altering and resending data packets (Integrity).
- Withstands Modification and replay attacks

Limitations of WPA

1. Older firmware and operating systems cannot be upgraded to support WPA.
2. WPA software must be present in computers, AP, and wireless adapters for WPA to function.
3. To connect multiple versions of WPA, the WPA and network clients must have the same configuration.
4. DoS and spoofing attacks can still happen even with WPA.
5. Transmission time is longer due to the additional packet size.

WPA 2

- More secure than WPA, because it uses a more advanced encryption key named "AES-CCMP."
- Helps wireless users in other ways, such as enabling them to disconnect from a wireless network and reconnect without having to go through a number of authentication process.
- **Limitations**
 1. Attackers have successfully compromised networks that use WPA and WPA2 encryption.
 2. Attackers can simply park near a location that broadcasts Wi-Fi and attempt to crack the network's security.
 3. WPA passwords are longer and more secure, but you should use WPA2 when you really need maximum protection.

WPA 3

- **In-transit Security:** Enhanced 128-bit encryption in WPA3-Personal and 192-bit encryption for WPA3-Enterprise implementations. *Using higher bit encryption significantly decreases the odds of compromising the key.*
- **Secure Authentication:** Uses a pre-shared key to join the network. Adds another layer of security or "handshake" called Simultaneous Authentication of Equals (SAE). The latest standard also introduces "forward secrecy", *which protects the ongoing communication even if the pre-shared key used to authenticate is compromised.*
- **Public Network Security:** Public networks such as in airports, malls and municipal networks are usually unencrypted or "Open." With WPA3, the communication over open network is automatically encrypted with a mechanism called Opportunistic Wireless Encryption (OWE). *This prevents eavesdropping while connected on a public network.*



Thanks